



SmoIVLM

Redefining Small and Efficient
Multimodal Models

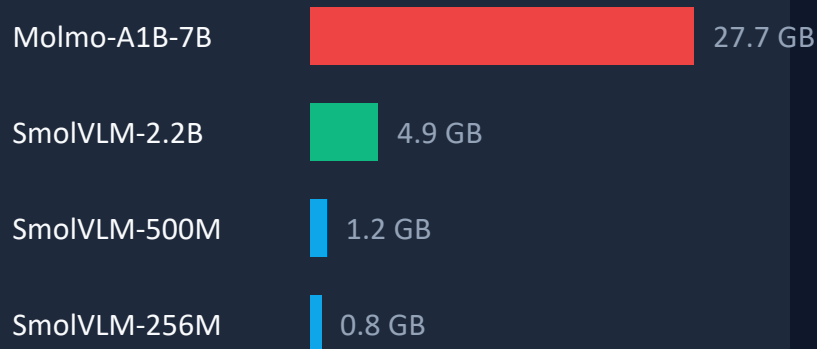
Hugging Face & Stanford University

Large VLMs Are Powerful but Impractical for Edge Deployment

The Edge Deployment Challenge

- State-of-the-art VLMs require massive compute.
- Cloud dependency adds latency and privacy risks.
- Mobile hardware lacks GPU memory for multi-gigabyte models.

Memory Footprint Comparison



SmolVLM Architecture Pipeline

Image Splitting, Pixel Shuffle, and SmolLM2 Backbone

1. Image Splitting

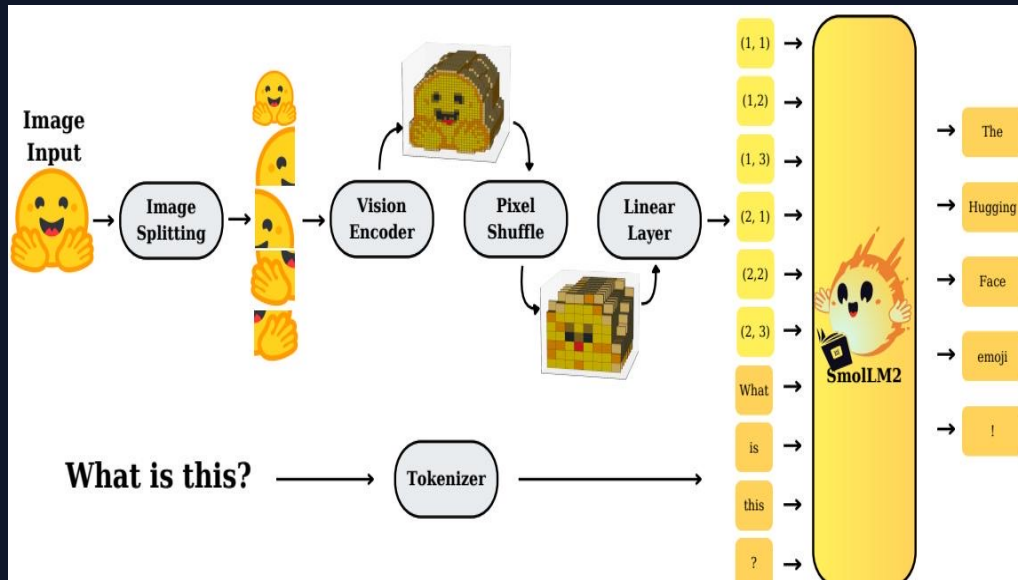
Images are split into sub-images for processing.

2. Vision Encoder & Shuffle

SigLIP vision encoder followed by aggressive pixel shuffle compression.

3. SmolLM2 Backbone

Tokens are concatenated with text embeddings into the LLM space.



Finding 1: Balanced Encoder-LM Compute

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Winner for Compact LMs (135M)

93M SigLIP-B/16

Provides the optimal balance of compute and performance for small models.

Only Justified at 1.7B LM Scale

428M SigLIP-SO400M

10% parameter overhead is not justified for small LMs, only for larger scales.

Finding 2: Aggressive Token Compression

Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Space-to-Depth Transformation

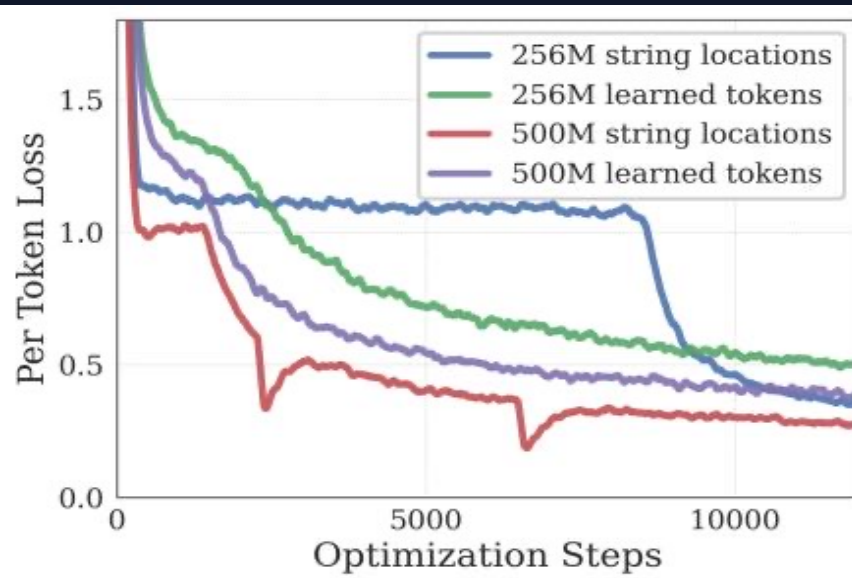
Rearranges spatial features into channels, effectively reducing sequence length.

Optimal for Compact VLMs

While large models use $r=2$, small models benefit from $r=4$ (16× reduction).

Impact

Eases attention overhead and significantly improves long-context modeling.



Finding 3: Context Length Scaling

Extending Context from 2K to 16K Tokens Unlocks Higher Resolution



RoPE Base Adjustment

Increased RoPE base from 10K to 273K to support extended sequences.

Data Mixing

Fine-tuned on a carefully curated mix of long and short-context data.

Model Capabilities

2.2B model runs at 16K context;
smaller 256M and 500M variants run
at 8K context.

Benchmark Performance

Competitive Performance at Sub-1GB to 4.9GB RAM

Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
RAM Usage	0.8 GB	1.2 GB	4.9 GB	27.7 GB (Molmo)
Average	44.0%	51.0%	59.8%	—
OCRBench	52.6%	61.0%	72.9%	54.7%
AI2D	46.4%	59.2%	70.0%	71.0%
ChartQA	55.6%	62.8%	68.7%	48.0%
TextVQA	50.2%	60.2%	73.0%	61.5%
DocVQA	58.3%	70.5%	80.0%	77.7%
ScienceQA	73.8%	80.0%	89.6%	87.5%
MMMU	29.0%	33.7%	42.0%	33.9%
MathVista	35.9%	40.1%	51.5%	37.6%
MMStar	34.6%	38.3%	46.0%	43.1%
Video-MME	33.7%	42.2%	52.1%	45.0%
* SmolVLM-256M surpasses the 300× larger ldefics-80B model. MLVU	40.6%	47.3%	55.2%	48.2%

Real-Time Inference on Edge Devices

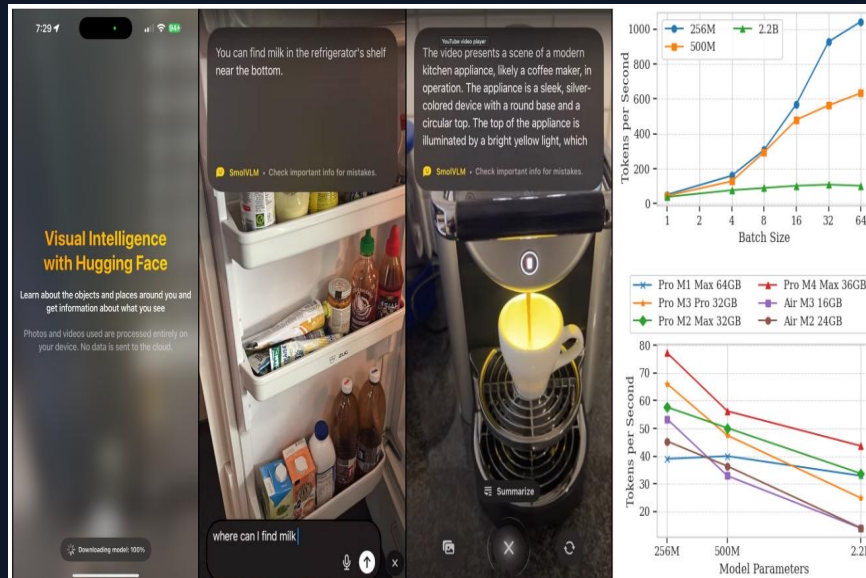
Up to 80 Tokens/Second on Apple Silicon via HuggingSnap App

100% Local Execution

Processes photos and videos entirely locally with no cloud dependency.

Throughput (SmoIVLM-256M)

- ~80 tokens/sec on Pro M1 Max 64GB
- ~55 tokens/sec on MacBook Air M3 16GB
- Nearly 1000 tokens/sec at batch size 64



Key Takeaways

Design Principles for Efficient Multimodal Models



Balance Encoder-LM Compute

Small VLMs benefit from proportionally smaller vision encoders.



Aggressive Token Compression

Pixel shuffle ($r=4$) is optimal for small models, reducing sequence length $16\times$.



Extend Context Length

Extending context to 8K-16K unlocks much higher image resolutions.



Sub-1GB Inference is Real

Architectural efficiency beats scale; on-device AI is viable today.

All models, data, and code are open-sourced to drive on-device AI research.

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